

Batch: DMW 2 Roll No.: 16010324044
Experiment/Assignment/Tutorial No.: 07
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Unsupervised learning using clustering algorithms

AIM: To learn k-means and hierarchical clustering algorithms for finding patterns in the data using unsupervised learning method

OUTCOME: Students will be able to apply the clustering algorithms to group similar data instances together

Procedure:

I] K-means clustering

1. Load and describe “Mall_Customers..csv” dataset
2. Plot Elbow plot and determine value of k for k-means clustering of iris dataset
3. Report the results of clustering using k-means algorithm

II] Hierarchical clustering

4. Load and describe “protein.csv” dataset
5. Plot the Dendrogram assuming 3 clusters for single, complete and average linkage methods
6. Report the result of clustering
7. Perform k-means and hierarchical clustering by exchanging the datasets and report your findings in detail

Observation Table:

Part A – K-means Clustering

Sr. No.	Observation / Analysis Question	Remark
1	Identify the number of records and features in the Mall Customers dataset. Which features are used for clustering?	
2	From the Elbow plot , identify the suitable value of K . Why did you select this value instead of a higher/lower K?	
3	After applying K-means, how many clusters are obtained? Briefly describe the characteristics of each cluster based on the feature values.	
4	From the 3D cluster plot , which clusters appear well separated and which clusters show overlap?	
5	Identify the approximate centroid/centre characteristics of each cluster. What does each cluster represent in the context of customers?	

1.) **Number of records:** 200

Number of features: 5

For clustering, the following **3 numerical features** are used:

- **Age**
- **Annual Income (k\$)**
- **Spending Score (1–100)**

2.) **K = 4**

K = 4 is selected because the WCSS value shows a noticeable reduction up to this point, after which the improvement becomes comparatively smaller.

- **Lower K values** may combine customers with different characteristics into the same cluster.
- **Higher K values** may create unnecessary extra clusters and make the model more complex.

Therefore, **K = 4** provides a good balance between clustering accuracy and simplicity.

3.)

Cluster	Approximate Characteristics	Customer Type
0	Age $\approx$ 45, Medium Income $\approx$ 49k, Medium Spending $\approx$ 43	Average/Moderate Customers
1	Age $\approx$ 33, High Income $\approx$ 87k, High Spending $\approx$ 82	Premium/High-Value Customers
2	Age $\approx$ 25, Low Income $\approx$ 29k, High Spending $\approx$ 74	Young High-Spending Customers
3	Age $\approx$ 40, High Income $\approx$ 87k, Low Spending $\approx$ 19	High-Income Low-Spending Customers

4.)

The **High Income–High Spending cluster (Cluster 1)** and **High Income–Low Spending cluster (Cluster 3)** appear relatively well separated because their spending behavior is significantly different.

The **Low Income–High Spending cluster (Cluster 2)** is also distinguishable due to its younger age and lower income.

The **Moderate customer cluster (Cluster 0)** may show some overlap with other clusters because its feature values lie closer to the middle range.

5.)

Part B – Cluster Validation using Silhouette Score

K	WCSS	Silhouette Score
2	212840.17	0.2932
3	143342.75	0.3839
4	104366.15	0.4055
5	75350.78	0.4443

1. **Which K gives the highest silhouette score?**

K = 5 gives the highest silhouette score (**0.4443**) among the tested values from 2 to 5.

2. **Which K was suggested by the Elbow method?**

From the provided code and experiment, the Elbow method suggests **K = 4**, which is also used in the K-Means clustering section.

3. **Are both methods suggesting the same K?**

No. The Elbow method suggests **K = 4**, while the highest silhouette score suggests **K = 5**.

4. **If they differ, which K would you select and why?**

I would select **K = 4** because the Elbow method identifies 4 as a good balance between reducing WCSS and avoiding unnecessary additional clusters. Although **K = 5** has a slightly higher silhouette score, **K = 4** provides a simpler and more interpretable clustering solution.

5. **What does the silhouette score tell you about the quality of the clusters?**

The silhouette score measures how well-separated and compact the clusters are. A higher silhouette score indicates that data points are closer to their own cluster and farther from other clusters, meaning better cluster quality. Scores closer to **1** indicate well-separated clusters, while scores near **0** indicate overlapping clusters. Here, the scores suggest **moderate clustering quality**, with $K = 5$ performing best among the tested values.

Part C – Hierarchical Clustering

Sr, No.	Observation / Analysis Question	Remark
1.	For the protein dataset, identify the number of records and features used for clustering. The Protein dataset contains 25 records and 9 numerical features used for clustering. Red Meat White Meat Eggs Milk Fish Cereals Starch Nuts Fr. Veg The Country column is used as an identifier and is not used as a clustering feature.	25 countries and 9 numerical features were used.
2.	Compare the single, complete and average linkage dendrograms. Which method produces comparatively well-separated clusters? Justify from the dendrogram. Among the three methods, Complete Linkage produces comparatively well-separated clusters. Single linkage tends to produce a chaining effect, where observations are progressively linked through nearby points. Complete linkage produces more compact clusters and shows clearer separation between major groups. Average linkage provides an intermediate structure between single and complete linkage.	Complete linkage gives comparatively clearer cluster separation.
3.	For 3 clusters, identify the groups obtained from the selected dendrogram. How are the observations grouped?	Three clusters

Sr, No.	Observation / Analysis Question	Remark
	Cluster 0: 22 observations Cluster 1: 1 observation Cluster 2: 2 observations So, single linkage produces one very large cluster and two small clusters, indicating a chaining effect. Here you should use the selected dendrogram from your actual output. For the single-linkage clustering actually implemented in your notebook, the cluster labels are:	were formed, but single linkage gives an uneven grouping.
4.	What happens to the clusters when the linkage method is changed? Explain with reference to the dendrograms. Single linkage can form elongated/chained clusters because it considers the closest pair of observations. Complete linkage tends to form more compact and well-separated clusters because it considers the farthest pair. Average linkage produces clusters based on average pairwise distances and generally gives an intermediate structure. Hence, changing the linkage method can change which observations belong to each cluster.	Cluster composition and separation depend on the selected linkage method.
5.	After exchanging the datasets, compare the suitability of K-means and hierarchical clustering for the two datasets. Which combination gives more meaningful clusters and why? K-Means on Protein dataset: successfully forms 3 clusters based on similarities in food-consumption features. Hierarchical clustering on Mall Customers: successfully forms clusters and provides dendrograms showing the hierarchical relationships between customers. However, the original combination of K-Means with Mall Customers and hierarchical clustering with Protein data gives more meaningful interpretations.	Algorithm suitability depends on the dataset structure and the interpretability of the resulting clusters.

Conclusion:

In this experiment, I successfully demonstrated K-Means and hierarchical clustering for unsupervised learning. K-Means was used to group Mall Customers into meaningful customer segments, while hierarchical clustering was applied to the Protein dataset using different linkage methods. The results showed that clustering can effectively identify patterns and groups in unlabeled data.

Signature of faculty in-charge with date